Estradiol, tamoxifen, and flaxseed alter IL-1β and IL-1Ra levels in normal human breast tissue in vivo.

INTRODUCTION: Sex steroid exposure increases the risk of breast cancer by unclear mechanisms. Diet modifications may be one breast cancer prevention strategy. The proinflammatory cytokine family of IL-1 is implicated in cancer progression. IL-1Ra is an endogenous inhibitor of the proinflammatory IL-1α and IL-1β. OBJECTIVE: The objective of this study was to elucidate whether estrogen, tamoxifen, and/or diet modification altered IL-1 levels in normal human breast tissue. DESIGN AND METHODS: Microdialysis was performed in healthy women under various hormone exposures, tamoxifen therapy, and diet modifications and in breast cancers of women before surgery. Breast tissue biopsies from reduction mammoplasties were cultured. RESULTS: We show a significant positive correlation between estradiol and in vivo levels of IL-1β in breast tissue and abdominal sc fat, whereas IL-1Ra exhibited a significant negative correlation with estradiol in breast tissue. Tamoxifen or a dietary addition of 25 g flaxseed per day resulted in significantly increased levels of IL-1Ra in the breast. These results were confirmed in ex vivo culture of breast biopsies. Immunohistochemistry of the biopsies did not reveal any changes in cellular content of the IL-1s, suggesting that mainly the secreted levels were affected. In breast cancer patients, intratumoral levels of IL-1β were significantly higher compared with normal adjacent breast tissue. CONCLUSION: IL-1 may be under the control of estrogen in vivo and may be attenuated by antiestrogen therapy and diet modifications. The increased IL-1β in breast cancers of women strongly suggests IL-1 as a potential therapeutic target in breast cancer treatment and prevention.